

SOP #10 – ATI (Analysis to Improve) for Emergency & Backup Generators

Goal: Ensure system readiness, reduce false starts, extend service intervals, and cut emission

1. Key Optimization Metrics

Metric	Ideal Value/Range	Optimization Tips
Monthly Load Test Runtime	30–60 mins under 30–50%	Avoid no-load runs; simulate critical loads
Battery Voltage Consistency	Stable within 10%	Use smart chargers and check monthly
Fuel Age & Quality	Fresh, clean (under 6 mos)	Use stabilizers, install polishing systems

2. Strategic Improvements

Automatic Battery Monitors: Alerts when voltage drops or aging occurs
 Fuel Polishing System: Prevents microbial growth in diesel fuel tanks
 Remote Monitoring & Telematics: Cloud alerts for faults and tests
 Thermal Imaging of Components: Identifies hotspots on alternator or coolant system
 Load Bank Optimization: Schedule real loads to prevent wet stacking

3. Legal, Safety & Compliance Considerations

NFPA 110 (Emergency Power Supply Systems): Covers performance/testing
 EPA Tier Regulations: Emissions limits for new and retrofit generators
 OSHA CFR 1910.147: Lockout/Tagout protocols for servicing
 NEC Article 700: Emergency systems installation and transfer switch specs
 Local Fire Code: Ventilation, exhaust stack distance, fuel storage location